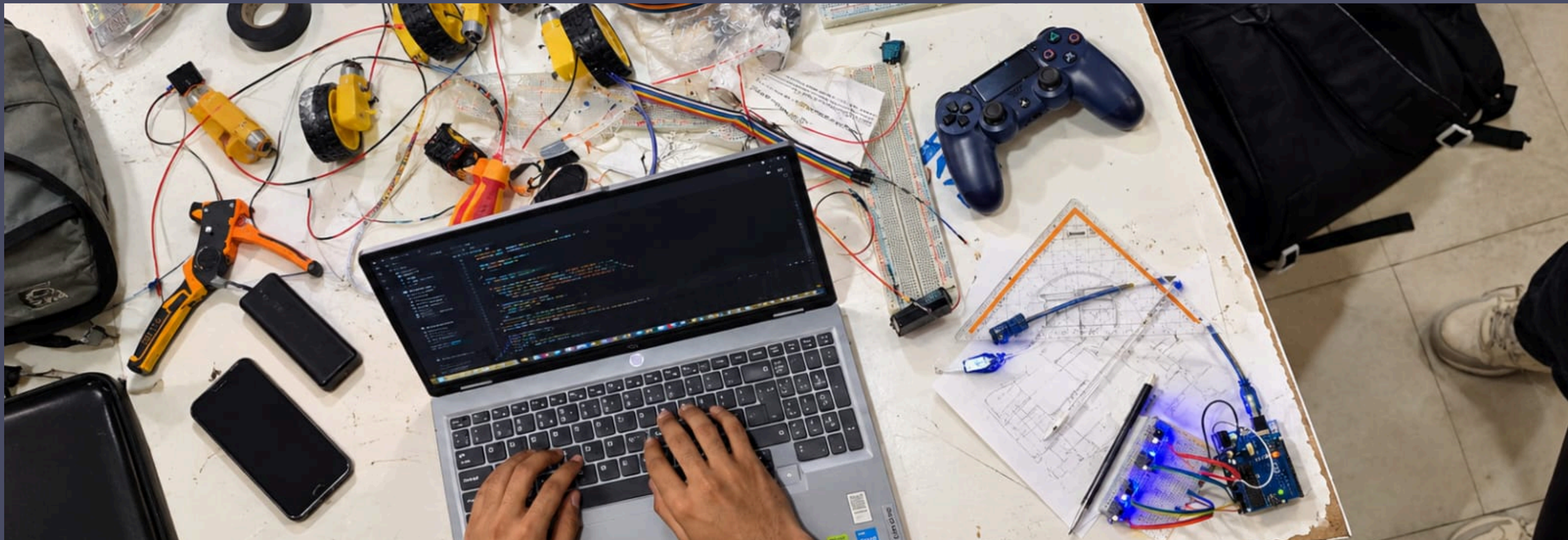


TEAM BASUONI

Arduino Track
DEEP DIVE 2026



Team members

ABDULLAH MOHAMMAD MESALAM

SHAHD MAMDOUH MAHMOUD

SARA AHMED ELMESALAMY

FARAH SABRI MOHAMMAD MADI

YOUSSEF ALI ABDULLAH ALI



UNITY IN INNOVATION



★ CONTENT ★

INTRODUCTION

SYSTEM OVERVIEW

AUTONOMOUS MODE

COMPONENT

OBJECTIVES

IMPLEMENTATION

CALCULATIONS

MANUAL MODE

INTRODUCTION

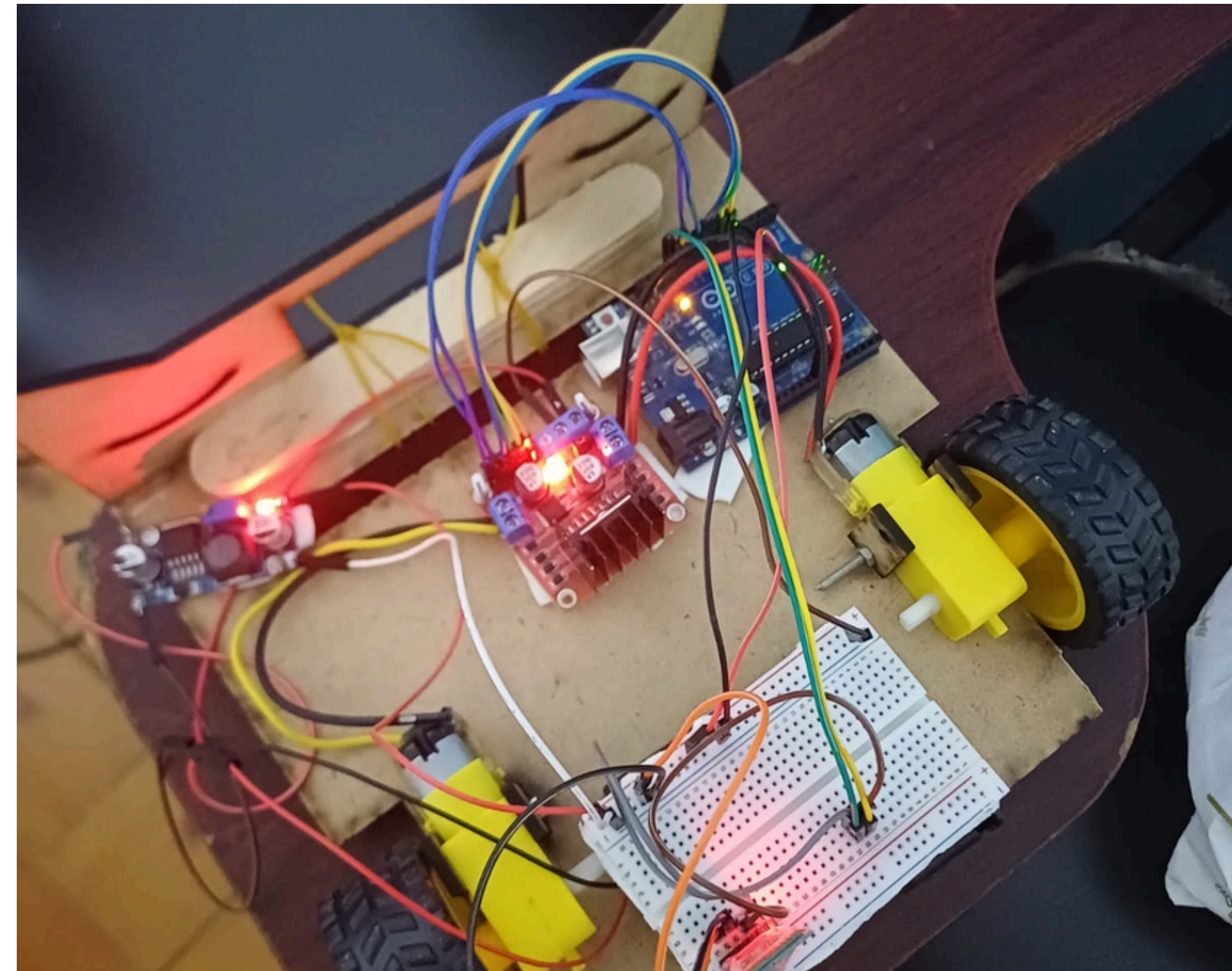
Our project is a robotic system designed to demonstrate the knowledge and skills we gained during the Embedded Systems training.

The robot operates in two different modes :

- Manual Mode
- Autonomous Line-Following Mode

The project combines hardware design, sensors, microcontroller programming, motor control, wireless communication, and autonomous control to build a complete robotic system.

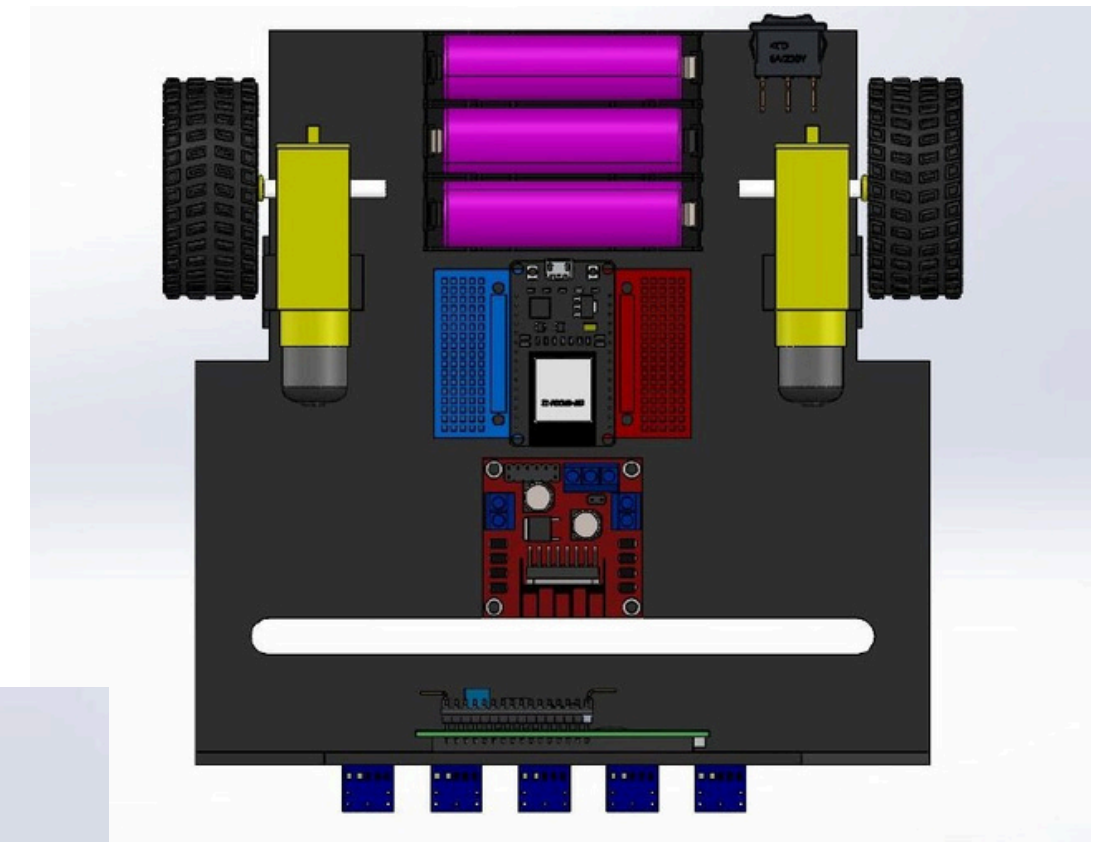
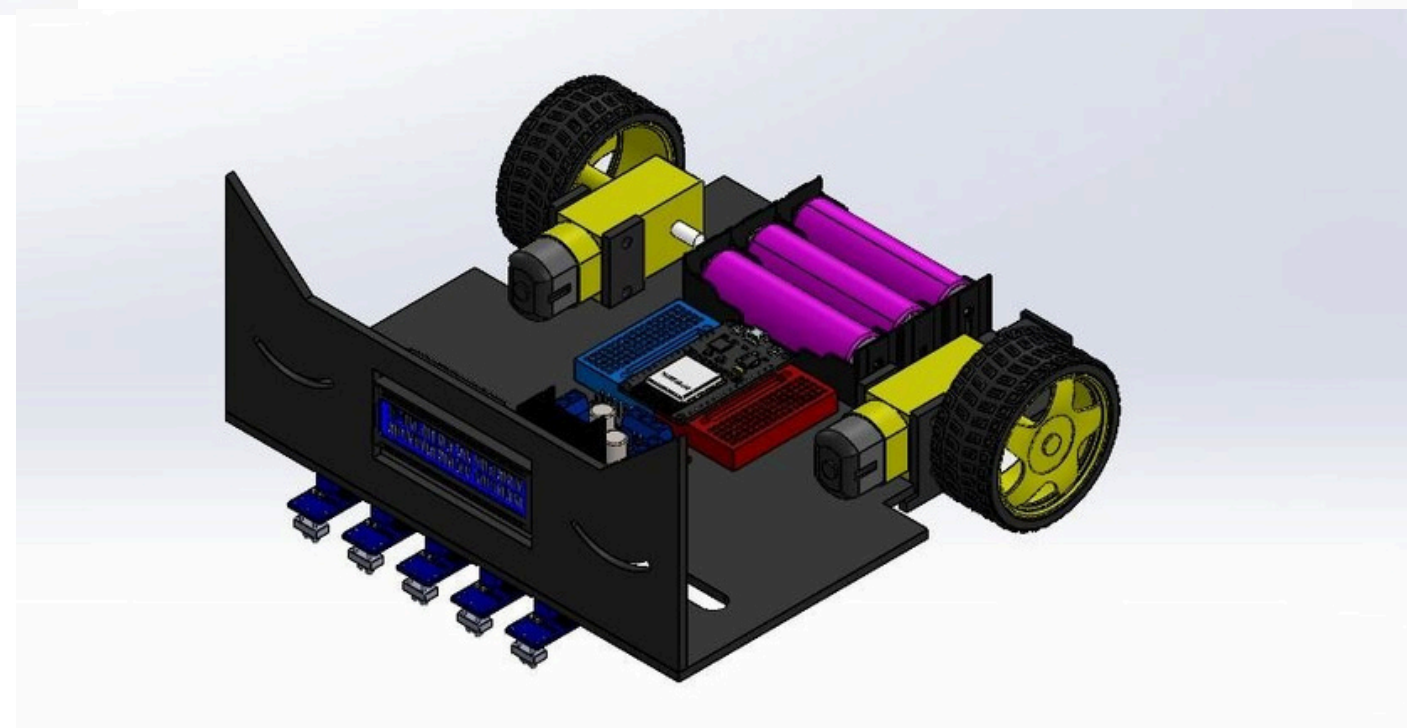
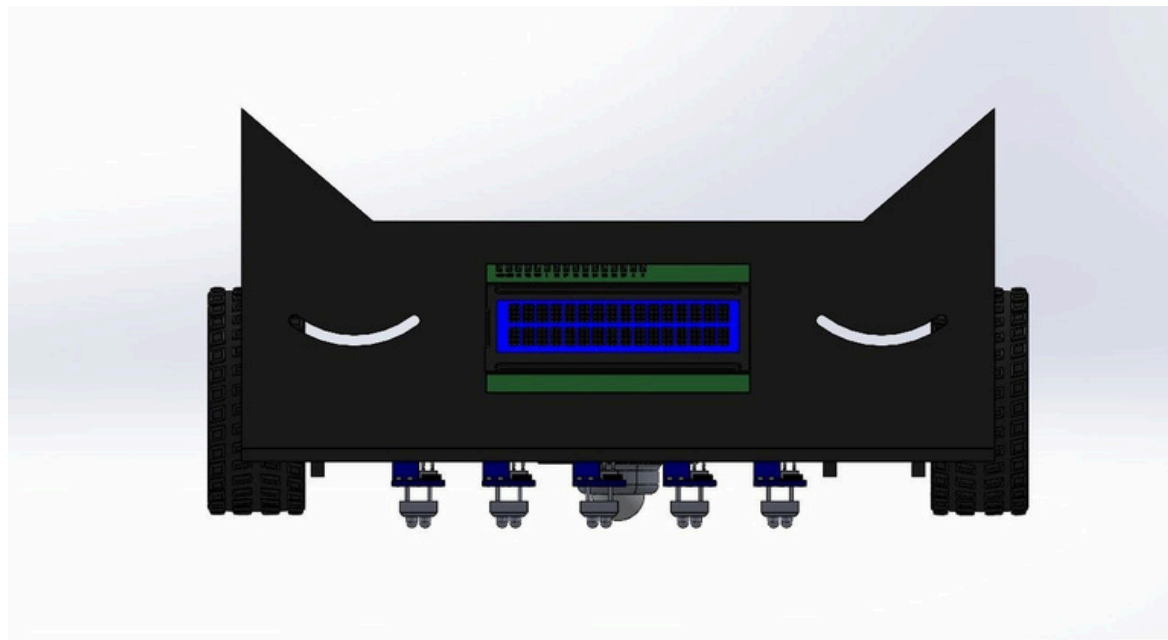
Our goal is to build a robot that can complete the track with high speed, accuracy, stability, and consistency.

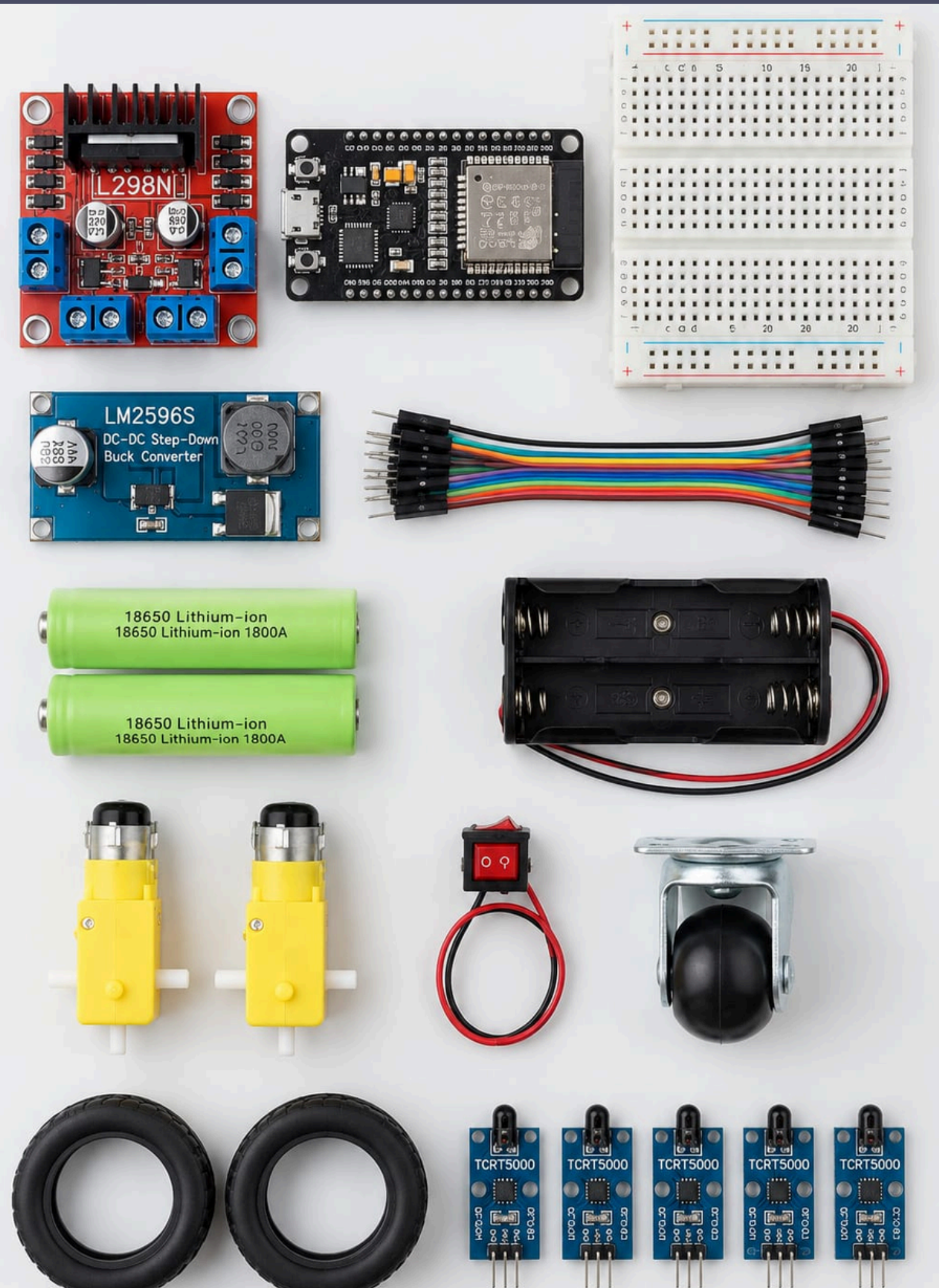


OBJECTIVES

- Design and integrate the electronic circuits
- Interface sensors with the microcontroller
- Control DC motors
- Implement PWM-based motor control
- Implement wireless communication
- Develop embedded software
- Develop an autonomous control algorithm
- Test, debug, and improve the complete system

MECHANICAL DESIGN





component selection

The components were selected according to the requirements of the robot and the two operating modes

ESP32

MOTOR DRIVER

DC MOTORS

BUCK CONVERTER

LCD

SWITCH

LITHIUM BATTERY

HOLDER

BREADBORD

WHEELS

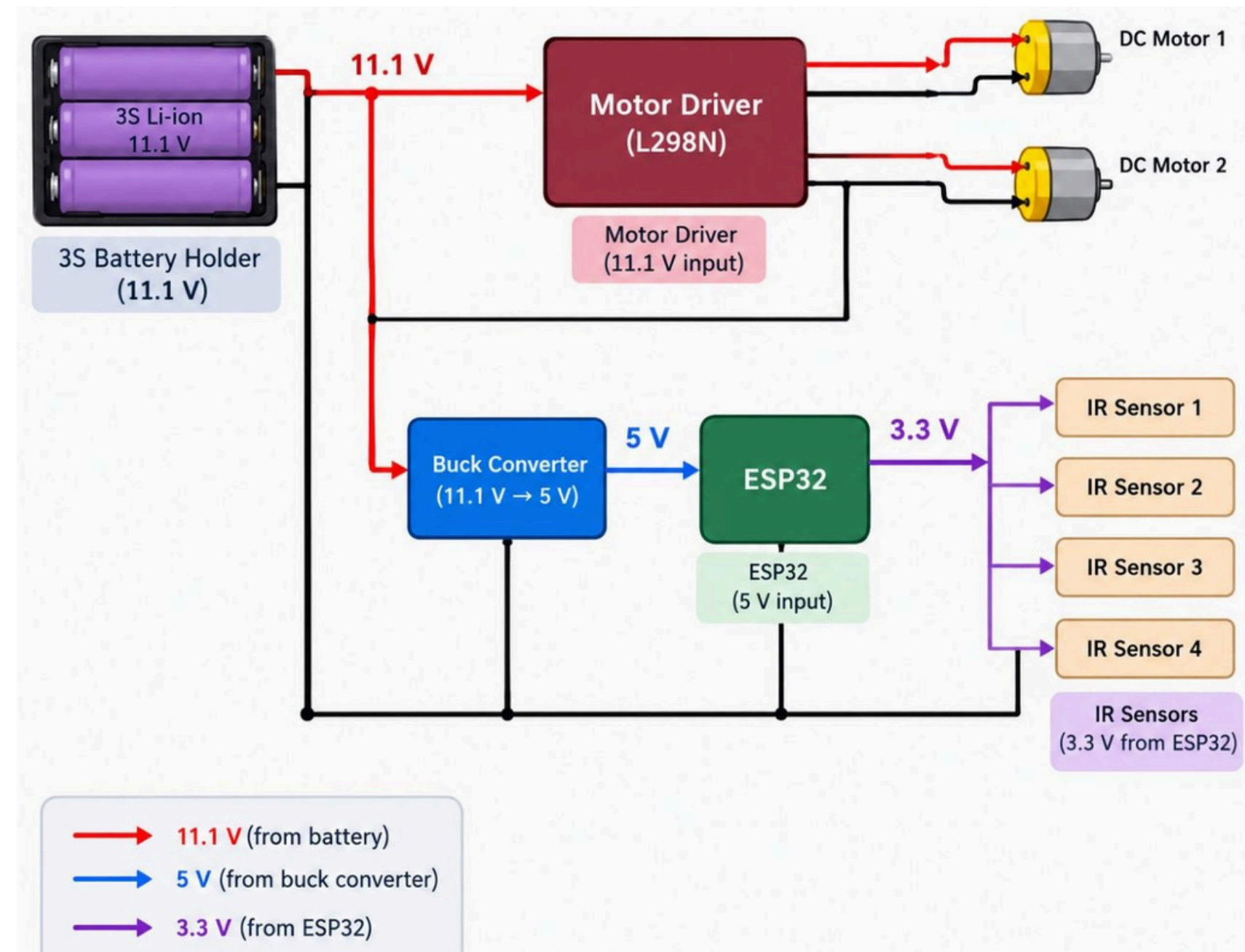
CASTERWHEEL

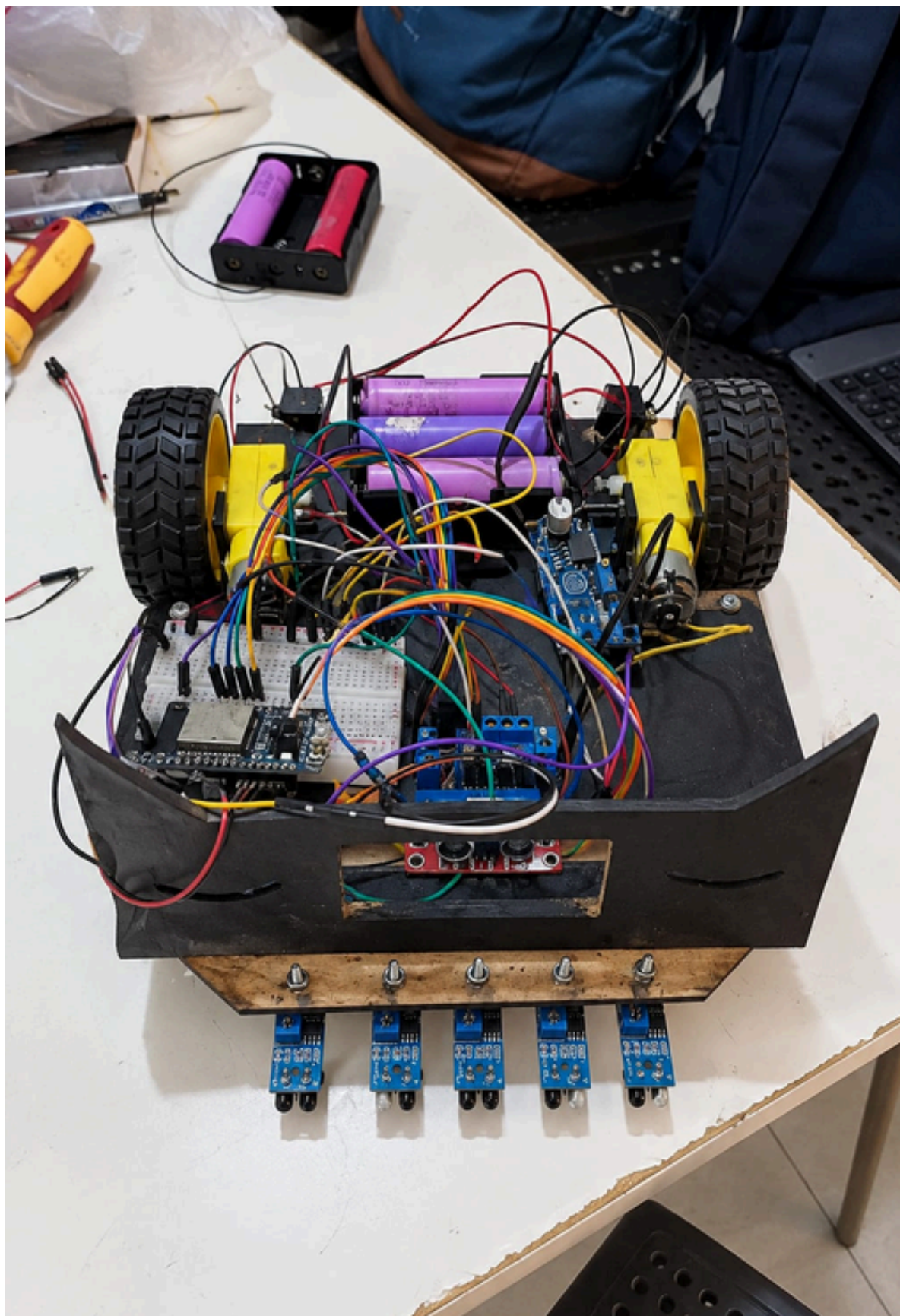
ANALOG IR SENSORS

POWER CALCULATIONS

Calculations :

The design calculations were used to make sure that the selected components and control parameters are suitable for the robot





IMPLEMENTATION

After designing the system, we implemented both the hardware and software parts of the robot.

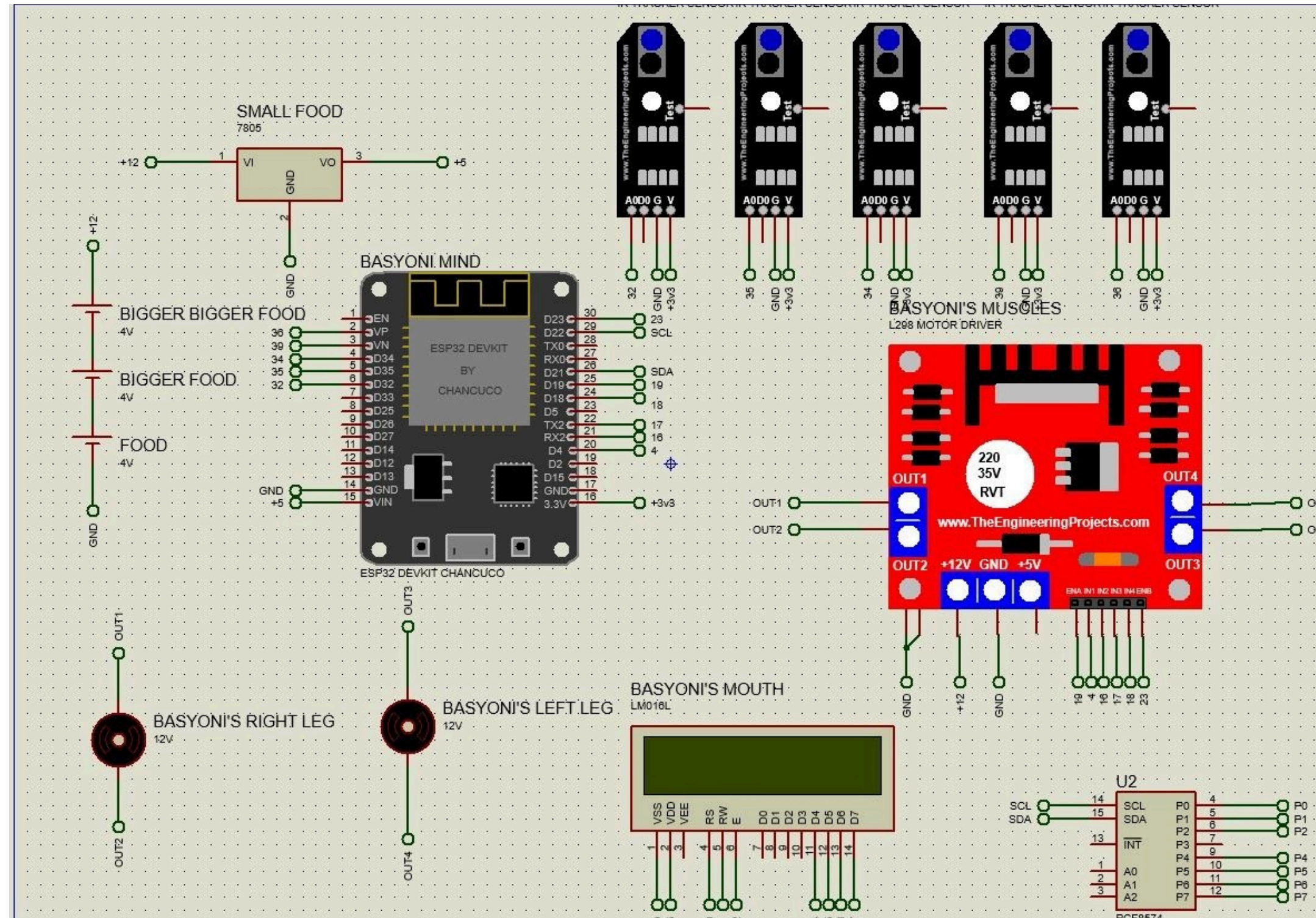
Hardware Implementation

- Assembling the robot chassis.
- Connecting the electronic components.
- Connecting the sensors and motors.
- Integrating the power system.

Software Implementation

- Programming the ESP32.
- Implementing Manual Mode.
- Implementing Autonomous Mode.
- Implementing the motor control logic.

SIMULATION



MANUAL MODE

CODE

In Manual Mode, the robot is controlled remotely by a team member using an approved wireless controller

```
if(cntrl->dpad() == 0x01) forward(speed);  
else if(cntrl->dpad() == 0x02) backward(speed);  
else if(cntrl->dpad() == 0x04) right(speed);  
else if(cntrl->dpad() == 0x08) left(speed);  
else if(cntrl->dpad() == 0x05) f_right(speed);  
else if(cntrl->dpad() == 0x09) f_left(speed);  
else if(cntrl->dpad() == 0x06) b_right(speed);  
else if(cntrl->dpad() == 0x0a) b_left(speed);  
else stop();
```

Motion

CODE

```
void forward(const int& speedr, const int& speedl) {
    analogWrite(EN1, speedr);
    analogWrite(EN2, speedl);
    digitalWrite(IN1, 0);
    digitalWrite(IN2, 1);
    digitalWrite(IN3, 0);
    digitalWrite(IN4, 1);
}

void backward(const int& speed) {
    analogWrite(EN1, speed);
    analogWrite(EN2, speed);
    digitalWrite(IN1, 1);
    digitalWrite(IN2, 0);
    digitalWrite(IN3, 1);
    digitalWrite(IN4, 0);
}
```

```
void right(const int& speed) {
    analogWrite(EN1, speed);
    analogWrite(EN2, speed);
    digitalWrite(IN1, 0);
    digitalWrite(IN2, 1);
    digitalWrite(IN3, 1);
    digitalWrite(IN4, 0);
}

void left(const int& speed) {
    analogWrite(EN1, speed);
    analogWrite(EN2, speed);
    digitalWrite(IN1, 1);
    digitalWrite(IN2, 0);
    digitalWrite(IN3, 0);
    digitalWrite(IN4, 1);
}
```

AUTONOMOUS MODE

CODE

In Autonomous Mode, the robot follows the designated line without human control

```
void pid_2() {
    const float weights[5] = {-2, -1, 0, 1, 2};

    int sensors[5];
    read_sensors_digital( sensors[0], sensors[1], sensors[2], sensors[3], sensors[4] );

    float sum = 0;
    float weighted = 0;
    int active = 0;

    // LOW = line detected
    for (int i = 0; i < 5; i++) {
        if (sensors[i] == LOW) {
            sum += 1.0;
            weighted += weights[i];
            active++;
        }
    }

    // All 5 sensors on line
    if (active == 5) {
        stop();
        return;
    }

    // No sensor sees the line
    if (active == 0) {
        if (!wasLost) {
            wasLost = true;
            lostSince = millis();
        }

        if (lastDir == 0) {
            stop();
            return;
        }
    }
}
```

```
float error = weighted / sum;

if (fabs(error) > 0.3f) {
    lastDir = (error > 0) ? 1.0f : -1.0f;
}

if (wasLost) {
    pidLastError = error;
    wasLost = false;
}

pidIntegral = constrain( pidIntegral + error, -50.0f, 50.0f );

float derivative = error - pidLastError;
pidLastError = error;

int correction =
    (int)(Kp * error +
        Ki * pidIntegral +
        Kd * derivative);

if (PID_DEBUG) {
    Serial.print(error);
    Serial.print(" | L=");
    Serial.print(base_speed + correction);
    Serial.print(" R=");
    Serial.println(base_speed - correction);
}

setMotors(
    base_speed + correction,
    base_speed - correction
);
}
```

SYSTEM OVERVIEW

The robot consists of three main parts :
Input → Processing → Output

Input

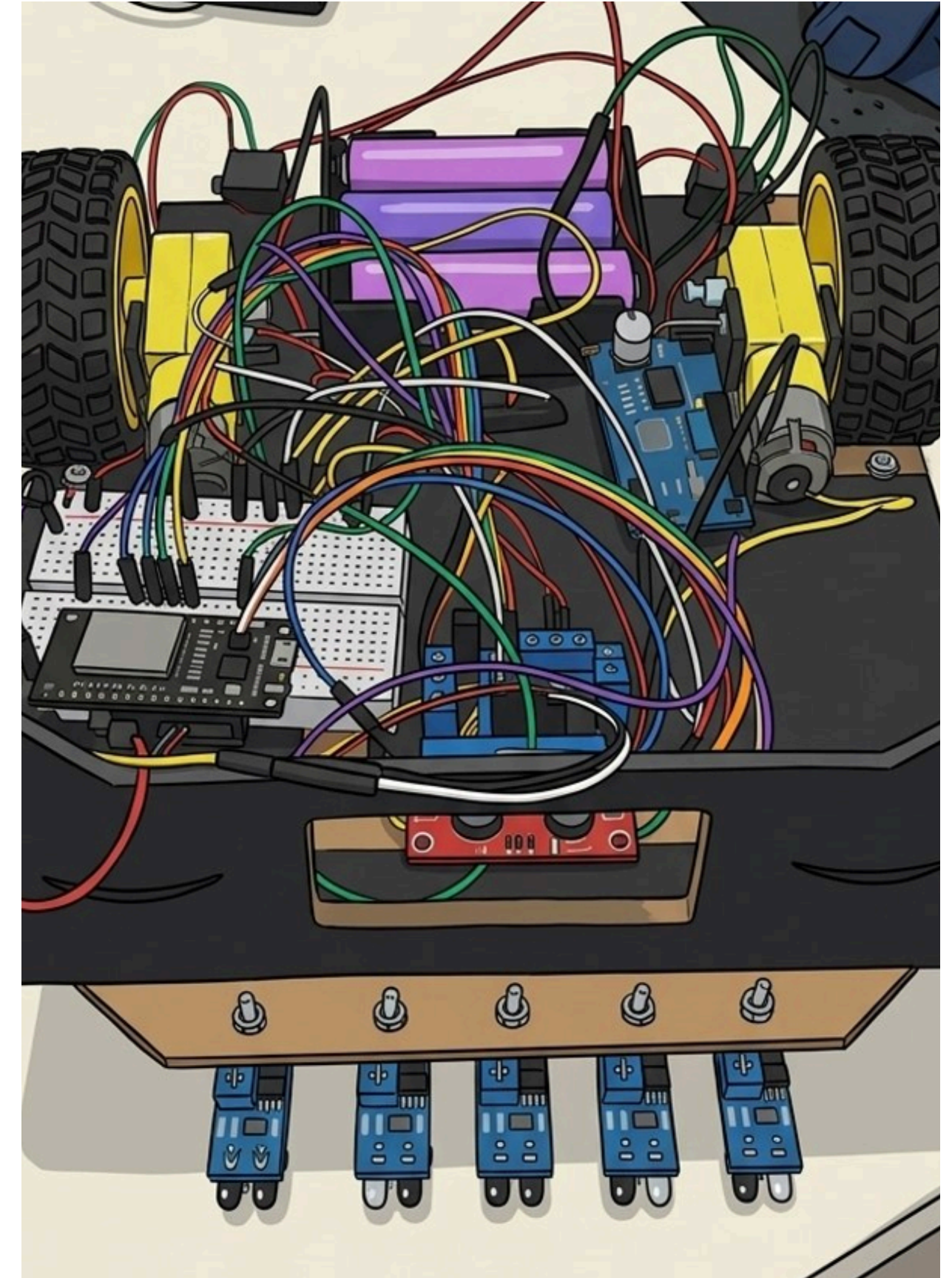
Sensors and wireless controller provide information to the system

Processing

The ESP32 receives the input, processes it, and makes the required decisions

Output

The motor driver controls the DC motors according to the ESP32 commands.



Thankyou

Arduino Track
DEEP DIVE 2026



— **TEAM BASUONI** —